ETALE-LOCAL PROPERTIES OF SCHEMES EXTEND UNIQUELY TO ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 0348: sources/stacks-chow/tags.txt, source lines 4083–4083. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{P}$ be a property of schemes which is local in the étale topology, see Descent, Definition 0348. The following are equivalent*

(1) *for some scheme U and surjective étale morphism $U \to X$ the scheme U has property $\mathcal{P}$, and*
(2) *for every scheme U and every étale morphism $U \to X$ the scheme U has property $\mathcal{P}$.*

If X is representable this is equivalent to $\mathcal{P}(X)$.

Proof. The implication (2) $\Rightarrow$ (1) is immediate. For the converse, choose a surjective étale morphism $U \to X$ with U a scheme that has $\mathcal{P}$ and let V be an étale X-scheme. Then $U \times_X V \to V$ is an étale surjection of schemes, so V inherits $\mathcal{P}$ from $U \times_X V$, which in turn inherits $\mathcal{P}$ from U (see discussion following Descent, Definition 0348). The last claim is clear from (1) and Descent, Definition 0348. □

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